
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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ABSTRACT

Efficient long-video understanding with Multimodal Large Language Models (MLLMs) requires more than reducing the number of visual tokens: the compressed token set must preserve both redundant stable content and the changes that drive events. Existing token pruning and merging methods improve efficiency mainly through importance or similarity, but this emphasis can underrepresent two reviewer-critical needs: joint spatio-temporal redundancy modeling and explicit preservation of turning points. We contribute ST-SimDiff, a training-free token selection framework built around the principle that similarity should identify redundancy while difference should protect key events. ST-SimDiff first constructs a spatio-temporal graph over video tokens to model spatial and temporal associations in a unified structure. It then performs two parallel selections: 1) similarity-based representative selection applies community detection to retain central tokens from redundant stable regions; and 2) temporal difference-based event selection detects sharp content changes and preserves tokens around dynamic transitions. This dual structure produces a compact subset that represents static context while retaining event-defining motion. Across extensive experiments, ST-SimDiff outperforms state-of-the-art efficient video understanding methods and substantially reduces computational cost. Our code will be released upon acceptance.

1 INTRODUCTION

Large Vision-Language Models (LVLMs) have advanced rapidly with the rise of Large Language Models (LLMs), showing strong performance on image and video understanding tasks (Zhang et al., 2024c; Liu et al., 2024c). For videos, however, the standard pipeline is costly: a video is sampled into frames, each frame is converted into hundreds or thousands of visual tokens, and the resulting sequence is passed to the LLM (Bai et al., 2025; Li et al., 2024). As video duration and resolution increase, this token sequence grows quickly, making long-video analysis and real-time interaction difficult because both computation and storage become prohibitive (Fu et al., 2024b). The central problem we address is therefore not whether tokens can be removed, but how to remove them while keeping the video evidence that matters.

Prior efficiency methods mainly expose two useful but incomplete contribution directions for LVLM token reduction. Importance-based pruning removes visual tokens with low model-derived importance; for example, FastV observes redundant attention distributions across layers and prunes low-attention visual tokens in deeper layers (Chen et al., 2024). Similarity-based merging or selection instead exploits repeated visual content within frames or across neighboring frames; FrameFusion merges similar tokens from adjacent frames (Fu et al., 2024b), while VisionZip selects dominant tokens at the vision-encoder level (Yang et al., 2025). These methods establish that many video tokens are removable, but they do not fully answer which tokens should survive when stable redundancy and abrupt temporal change coexist.

The limitation of existing token reduction methods is twofold, and both limitations directly motivate our contribution. First, many approaches model either spatial correlations within a frame or temporal correlations at fixed positions across frames, rather than jointly analyzing spatio-temporal relationships among video tokens. This separation weakens their ability to identify complex redun-

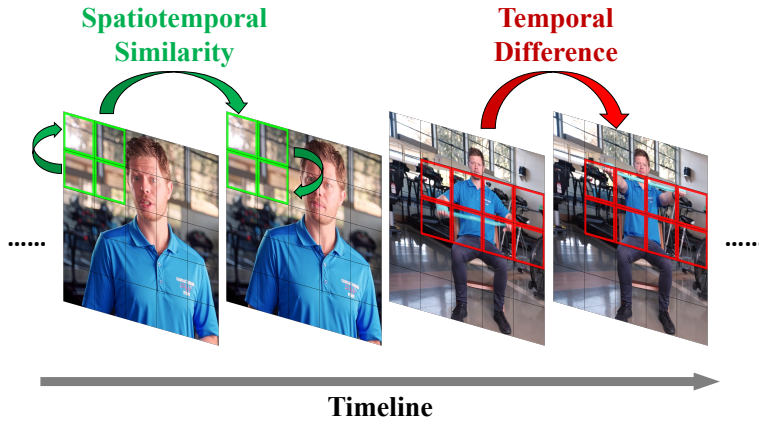


Figure 1: Contribution motivation of ST-SimDiff. Efficient video understanding must address two complementary token-selection signals: Spatiotemporal Similarity (left) supports compression of redundant content within frames and across adjacent frames, while Temporal Difference (right) preserves the actions and event transitions that define the video narrative.

dancies that arise from persistent objects, backgrounds, and motion patterns. Second, the dominant compression signals are commonality-oriented—importance and similarity—so the methods can overlook rare but decisive changes. Because a video’s meaning is often determined by turning events, a compression rule that only rewards similarity may smooth over sudden actions and remove evidence needed for correct interpretation.

We propose **ST-SimDiff** to make this missing structure explicit: similarity is used to find redundancy, and difference is used to preserve key moments. The framework is built on a spatio-temporal graph whose nodes are all video tokens and whose edges encode spatial and temporal token relations. On this graph, ST-SimDiff performs **similarity-based representative token selection**, grouping highly related tokens through community detection and retaining central tokens from each community to represent stable visual content. In parallel, it performs **difference-based event token selection**, inspecting temporal edges for sharp drops in similarity and preserving tokens that mark content-changing transitions. The final token set is the union of these representative and event tokens, giving the LLM a compact input that keeps both the stable context and the dynamic evidence.

Our contributions are organized as follows:

- We introduce a similarity–difference perspective for efficient video understanding, arguing that VLM token compression should jointly value redundant stable content and event-defining change.
- We design ST-SimDiff, a training-free spatio-temporal graph framework that models video-token relationships uniformly and combines similarity-based representative token selection with difference-based event token selection.
- We validate ST-SimDiff on long-form video understanding benchmarks, showing that it substantially reduces the visual token budget while maintaining or improving performance over state-of-the-art compression methods.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) have enabled a broad family of Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). A typical LVLM converts images or video frames into visual tokens with a pre-trained visual encoder, then aligns those tokens to an LLM through modules such as MLP projectors (Liu et al., 2024a;b) or Q-Formers (Li et al., 2023; Dai et al., 2023). This architecture is the foundation

for strong multimodal reasoning capabilities (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a), including recent progress on longer and higher-resolution video understanding (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

The same architecture also creates the efficiency bottleneck that our work targets. Video inputs require continuous frame sequences, so their visual-token counts are much larger than those of static images and grow with both video length and resolution. This token growth makes training and inference expensive, especially for long-video understanding. Our contribution is positioned within this bottleneck: we seek a token subset that reduces computation while preserving both stable visual context and dynamic event evidence.

2.2 VISUAL TOKEN COMPRESSION

Visual token compression is a direct response to the high cost of video LVLM inference. Existing methods can be read as two main lines of evidence for what makes tokens removable: importance and similarity. Importance-based methods prune tokens estimated to contribute less to model behavior, often using attention scores or related feature metrics; representative examples include FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). Similarity-aware methods further recognize that tokens may be individually useful but mutually redundant, and therefore merge or select tokens to reduce repeated information (Shang et al., 2024; Ye et al., 2025). FrameFusion (Fu et al., 2024b) combines similarity-based merging with importance-based pruning, VisionZip (Yang et al., 2025) selects dominant tokens and merges contextual information, and VisPruner (Zhang et al., 2025b) removes redundancy using visual cues.

These advances motivate our design but also clarify the gap addressed by ST-SimDiff. Most similarity-driven video methods do not explicitly model the global spatio-temporal similarity structure among tokens, even though redundancy appears both within frames and across time. Methods such as FrameFusion mainly exploit temporal similarity between adjacent frames, which limits their ability to use richer spatio-temporal associations under extreme token reduction. More importantly, the field has largely optimized for preserving common or representative information, while giving less explicit attention to difference signals that mark event transitions. ST-SimDiff contributes a complementary formulation by jointly compressing similarity-based redundancy and retaining difference-based event tokens.

3 PROBLEM DEFINITION

We formulate efficient video understanding as a reviewer-facing token selection problem. Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) encodes the video into a full sequence of N visual tokens, $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. Because self-attention scales quadratically with the token count N , i.e., $O(N^2)$, directly passing all video tokens is computationally expensive. Our goal is to learn no new parameters, but to design an efficient token selection function, $f(\cdot)$, that maps the full sequence T_{full} and compression ratio r to a compact subset $T_{\text{sub}} = f(T_{\text{full}}, r)$. The subset should keep $|T_{\text{sub}}|$ approximately $r \cdot N$ while maximizing downstream LVLM performance. Thus, the key contribution of the method is the selection criterion: $f(\cdot)$ must preserve both representative stable content and critical temporal changes within T_{sub} .

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As illustrated in Figure 2, ST-SimDiff makes the paper’s central contribution operational through two parallel selectors over one spatio-temporal token graph: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). SRTS contributes the redundancy-compression path. It applies community detection to group highly similar tokens, such as persistent background or stable objects, and retains central tokens from each community to form the representative set T_{rep} . DETS contributes the event-preservation path. It inspects temporal graph edges and identifies adjacent-frame positions where similarity drops sharply; these high-difference locations are treated as turning points, and their tokens are retained as event tokens T_{trans} . The final

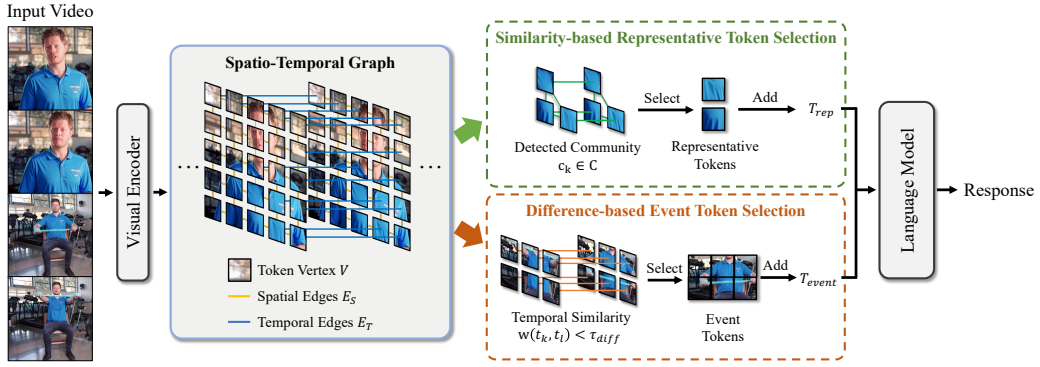


Figure 2: ST-SimDiff contribution structure. The framework first constructs a spatio-temporal graph, then selects complementary token sets through Similarity-based Representative Token Selection and Difference-based Event Token Selection.

subset $T_{\text{sub}} = T_{\text{rep}} \cup T_{\text{trans}}$ is therefore compact by construction but remains event-aware, because stable content and dynamic transitions are selected by distinct, complementary criteria.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

The first technical contribution is a unified graph representation for video tokens. Given a video, the vision encoder produces tokens $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ has a feature vector $\mathbf{x}_k$ and a precise spatio-temporal location, specified by its frame index $T(t_k)$, spatial height index $H(t_k)$, and spatial width index $W(t_k)$. This notation lets the same structure express both within-frame neighborhoods and across-frame continuity.

We model these relationships with a spatio-temporal graph $G = (V, E)$, where V contains all tokens in T . The edge set is decomposed as $E = E_S \cup E_T$, making the contribution of the graph explicit: E_S captures local spatial adjacency, while E_T captures temporal continuity. The spatial edges E_S connect tokens that are adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges E_T connect tokens at the same spatial position in adjacent frames, providing the structure later used to detect event-defining changes:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

Each graph edge is weighted by feature similarity, so the same representation can support both similarity-based compression and difference-based event detection. Specifically, for any edge $w(v_i, v_j) \in E$, the weight is the cosine similarity between the corresponding token features:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph is the shared substrate for the full method: it encodes local spatial relations, temporal continuity, and edge weights that can be interpreted either as evidence of redundancy or as evidence of change.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content contains substantial spatio-temporal redundancy, and SRTS is the component that turns this observation into an explicit compression contribution. In a static scene, background tokens remain similar across time; for a moving object, semantically related tokens may shift spatial positions while still representing the same visual entity. Rather than treating these cases separately,

SRTS uses graph communities to find redundant token groups across the spatio-temporal structure. Within the graph, semantically related tokens form dense “communities” or “clusters” based on feature similarity, regardless of their exact frame or location. Selecting representatives from each community therefore compresses repeated information while preserving the core semantics of the video.

To accurately identify strongly correlated token clusters, we first threshold the graph G . We set a similarity threshold τ_{sim} and retain only the edges with weights above this threshold, forming a new graph $G' = (V, E')$, where $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. Next, we apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) on G' to identify token clusters $C = \{c_1, c_2, \dots, c_m\}$. For each community $c_k \in C$, we rank and filter its internal tokens based on their centrality. The centrality score $S_c(t_a)$ of a token $t_a \in c_k$ is defined as its average similarity with all other tokens within the community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

Subsequently, we set the intra-community retention rate to the globally preset compression ratio r , and preserve the top $\lceil |c_k| \cdot r \rceil$ tokens with the highest centrality scores from each community c_k . This uniform filtering process naturally handles all communities, including single-node communities composed of unique tokens (in which case the sole node is retained). Formally, the set of representative tokens, T_{rep} , is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score}=S_c(t)}(\lceil |c_k| \cdot r \rceil) \quad (5)$$

where the TopK function returns the set of k elements with the highest scores from a given set. The resulting contribution of SRTS is a compact representation of stable and persistent video content: each community contributes only its most central tokens, reducing redundancy while preserving the semantic coverage of the cluster.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

DETS provides the complementary contribution to SRTS: it protects the changes that similarity-based compression can miss. If similarity describes the “norm” of a video, difference describes its “events”. A selector that only preserves what is common may represent “what is” while losing evidence of “what happened”. Events such as a new object appearing, an action beginning, or a scene transition are expressed as sharp feature differences between temporally adjacent tokens. Detecting these turning points is therefore essential for retaining dynamic evidence needed for questions about timing, causality, and action.

We implement this event-preservation path by analyzing the temporal edges (E_T) in the spatio-temporal graph. For any temporal edge $(v_k, v_l) \in E_T$ connecting adjacent tokens t_k and t_l , a large temporal difference indicates a candidate event boundary, such as object entry, scene change, or action start/end. We use a dynamic threshold τ (e.g., the 95th percentile of all temporal edge difference scores) and retain the later token t_l whenever the temporal similarity falls below the difference threshold, i.e., $w(t_k, t_l) < \tau_{\text{diff}}$. Formally, the event-token set T_{event} is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

This rule makes the difference contribution visible in the final token set: moments with significant temporal change are preserved even when they are not representative of the surrounding content. As a result, compression does not discard the dynamic transitions that often determine the video’s narrative.

4.5 OVERALL REDUCTION PROCESS

The overall reduction process exposes how the two contributions are combined under a fixed budget. We compute the representative token set T_{rep} and event token set T_{event} in parallel, then form the

initial candidate set $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$. To meet the target count $N_{\text{target}} = \lceil r \cdot N \rceil$, we apply a final pruning step only when $|T_{\text{candidate}}|$ exceeds the budget. In that case, we remove the $|T_{\text{candidate}}| - N_{\text{target}}$ tokens with the lowest importance. Following existing work (Chen et al., 2024), token importance is measured by the shallow-layer attention score of the LLM. This final step keeps ST-SimDiff budget-flexible while preserving the similarity-selected representatives and difference-selected event evidence as much as possible.

In summary, ST-SimDiff combines graph-based token selection with attention-based budget adjustment. The graph stage is the main contribution: it uses similarity to identify representative redundancy and difference to preserve event transitions. The attention stage then adapts the candidate set to the requested compression ratio. This design keeps the compression process principled, budget-aware, and aligned with the paper’s central balance between efficiency and video-event fidelity.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The efficiency contribution of ST-SimDiff is supported by the complexity of its three stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

Graph construction requires computing cosine similarity only between spatially and temporally adjacent neighbors. Because each of the N tokens has a small constant number of such neighbors, this stage is a single pass over the tokens with complexity $O(Nd)$, where d is the feature dimension.

After the graph is built, SRTS identifies token communities. As detailed in our implementation (Section 5.2), we use connected components, whose near-linear time complexity is $O(N + |E|)$. Since $|E|$ is at most $3N$, this reduces to $O(N)$. The within-community filtering step could otherwise dominate the cost: a naive all-pairs computation inside each community c_k would require $O(|C| \cdot |c_k|^2 \cdot d)$. To keep the representative-selection contribution efficient, we cap the maximum community size and partition any rare community larger than a threshold such as $|c_k| > \sqrt{N}$. Although Louvain and Leiden can provide more complex community definitions, their complexity approaches $O(N \log N)$, so connected components better matches our goal of minimal processing overhead.

DETS is also lightweight: it scans temporal edges once and compares their similarities with the difference threshold to identify significant changes. This event-selection path therefore has linear complexity $O(Nd)$.

Overall, ST-SimDiff is governed by linear-time operations, yielding total complexity $O(Nd)$. This overhead is substantially lower than the LLM self-attention cost, $O(N^2d)$. The method therefore contributes token reduction without introducing a computational bottleneck that would offset the efficiency gains.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines The experiments are organized to test three claims about ST-SimDiff: whether it improves compressed-token accuracy, whether the gain transfers across base models, and whether its two selectors each contribute. We use LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c) as base models and compare against state-of-the-art video token compression techniques. The baselines cover **importance-based reduction methods**, including FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods**, including VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). We also include the original uncompressed model (Vanilla) as a performance upper bound and Random Sampling with the same token count as a lower bound.

Benchmarks We evaluate on three benchmarks that make the contribution measurable under long-video reasoning demands: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME covers diverse video types and reports Short, Medium, and Long categories, allowing us to inspect performance across video duration.

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Main contribution evidence on LLaVA-Video-7B with 64 input frames (%). ST-SimDiff is compared with efficient token compression baselines on long-form video understanding benchmarks, and the best result among compressed methods is emphasized in **bold**.

LongVideoBench evaluates retrieval and reasoning over long videos through a “referring reasoning” task, with videos of 15s, 60s, 600s, and 3600s. EgoSchema focuses on long-term first-person question answering, where character intentions and causal action chains make event preservation especially relevant.

5.2 IMPLEMENTATION DETAILS

All implementation choices are fixed to make the efficiency and contribution comparisons direct. We run inference on NVIDIA L20 GPUs with 48GB VRAM under Ubuntu 22.04 and use the *lmms-eval* (Zhang et al., 2024a) library. LLaVA-Video follows its original 64-frame setting, and NVILA is also evaluated with 64 input frames so that resource usage is comparable. We set the similarity threshold to $\tau_{\text{sim}} = 0.8$ and the difference threshold to $\tau_{\text{diff}} = 0.2$, and report token retain ratios of $r = 30\%$ and $r = 50\%$. For the practical community-detection step, we use connected component finding to balance performance improvement with low additional time cost.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

Tables 1 and 2 provide the main evidence for the paper’s contribution: ST-SimDiff improves compressed video understanding by selecting both representative and event-sensitive tokens. We report results on VideoMME, LongVideoBench, and EgoSchema for two base models, LLaVA-Video-7B and NVILA-8B, at retain ratios of 30% and 50%. This design tests both high-compression behavior and robustness across model backbones.

(1) The first result is that ST-SimDiff consistently outperforms competing compressed-token methods across the tested configurations. At a 50% token retain ratio, the overall performance on both base models surpasses other compression algorithms and, on some benchmarks, matches or exceeds the original model using 100% of the tokens. This result supports the central claim that balancing

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Generalization evidence on NVILA-Video-8B with 64 input frames (%). The comparison tests whether the ST-SimDiff contribution transfers across base models, with the best compressed-method result emphasized in **bold**.

similarity and difference is more effective than compressing tokens through redundancy-oriented criteria alone.

(2) The baseline pattern clarifies why ST-SimDiff contributes beyond prior strategies. Among importance-based methods such as FastV and FasterVLM, FasterVLM is usually stronger, but importance scores can still retain tokens that are important yet repetitive, leaving temporal redundancy under-compressed. Hybrid methods such as PruMerge, VisionZip, and FrameFusion are more competitive, and FrameFusion often gives the strongest baseline results. Even so, these methods primarily optimize for representative “commonality” information, which leaves a blind spot for narrative evidence carried by “turning points” and “abrupt events”.

(3) Overall, the comparison shows that existing baselines increasingly understand video “redundancy” but still leave two gaps: they do not fully analyze joint spatio-temporal similarity between visual tokens, and they do not explicitly select for “difference” signals that define plot development. ST-SimDiff’s gains are consistent with addressing both gaps through a spatio-temporal graph plus explicit event-token selection.

5.4 ABLATION STUDY

Table 5 isolates the two technical contributions of ST-SimDiff. Starting from a Baseline that uses only fundamental importance-based pruning, we add the similarity selection module (+Sim) and then the difference selection module (++Diff). The ablation is evaluated at retain ratios of 30% and 50%, so the results show not only whether each module helps, but also where its contribution is most visible under different compression strengths.

(1) The +Sim rows show the contribution of representative similarity modeling. Adding similarity selection consistently improves over the baseline, and the comparison among spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.) variants shows that joint spatio-temporal modeling is the most effective way to compress redundancy.

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Component contribution ablation on LLaVA-Video-7B (%). +Sim tests similarity-based representative selection, and ++Diff tests the added value of difference-based event selection.

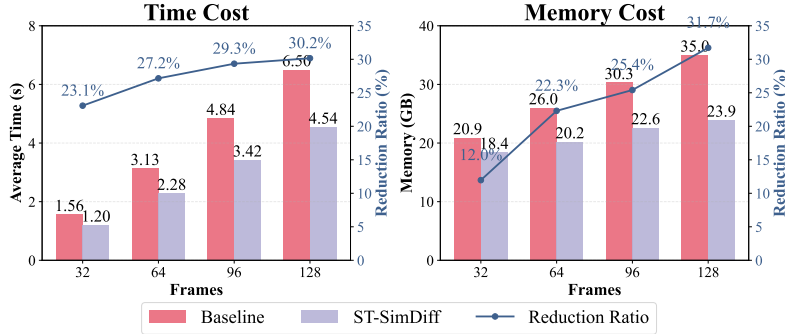


Figure 3: Efficiency contribution of ST-SimDiff under a 30% token budget. Average inference time and peak GPU memory are compared with the uncompressed LLaVA-Video baseline across input lengths.

(2) The ++Diff rows show the added contribution of event-token preservation after the best joint similarity module is already included. The improvement is especially clear at the more aggressive 30% retain ratio, where similarity-based compression has the greatest risk of dropping rare but narrative-critical tokens. At 50%, the marginal gain is smaller because the larger retained set is more likely to include these event tokens already and performance is closer to its upper bound. This ablation directly supports the paper’s core structure: similarity compresses redundancy, while difference preserves the transitions that make the compressed representation reliable.

5.5 COMPUTATIONAL COST ANALYSIS

Figure 3 evaluates the efficiency side of the contribution. Under a 30% token budget, we compare ST-SimDiff with the uncompressed LLaVA-Video baseline in average inference time and peak GPU memory. The tested input lengths range from 32 to 128 frames, showing how the method behaves as video inputs grow.

Inference Time ST-SimDiff turns token reduction into measurable speedup. As the number of input frames increases, the uncompressed baseline grows from 1.56 seconds at 32 frames to 6.50 seconds at 128 frames. ST-SimDiff slows this growth, reaching 4.54 seconds at 128 frames. The

time-saving rate therefore increases with video length, from 23.0% at 32 frames to 30.2% at 128 frames.

Memory Cost ST-SimDiff also reduces the memory bottleneck created by long token sequences. The baseline peak GPU memory increases from 20.9 GB at 32 frames to 35.0 GB at 128 frames, while ST-SimDiff lowers the number of tokens passed to the LLM. For 128-frame videos, peak memory is 23.9 GB, corresponding to a 31.7% memory saving relative to the baseline. This result supports the practical value of the contribution for longer videos and memory-limited hardware.

In summary, the experiments show both sides of the contribution: ST-SimDiff improves compressed-token accuracy and reduces the time and memory costs that make long-video LVLM inference difficult.

6 CONCLUSION

We presented ST-SimDiff as a training-free framework for efficient long-video understanding with MLLMs. The paper’s main contribution is a clearer token-selection principle: similarity should be used to compress redundant stable content, while difference should be used to preserve event-defining transitions. ST-SimDiff implements this principle with a spatio-temporal graph, similarity-based representative token selection through community structure, and temporal difference-based event token selection at key transition points. Experiments on VideoMME, LongVideoBench, and EgoSchema show that this structure improves over state-of-the-art compression methods while reducing inference time and peak GPU memory. These results support a reviewer-facing takeaway: effective video-token compression should balance content representativeness with transitional events rather than optimizing for redundancy alone.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

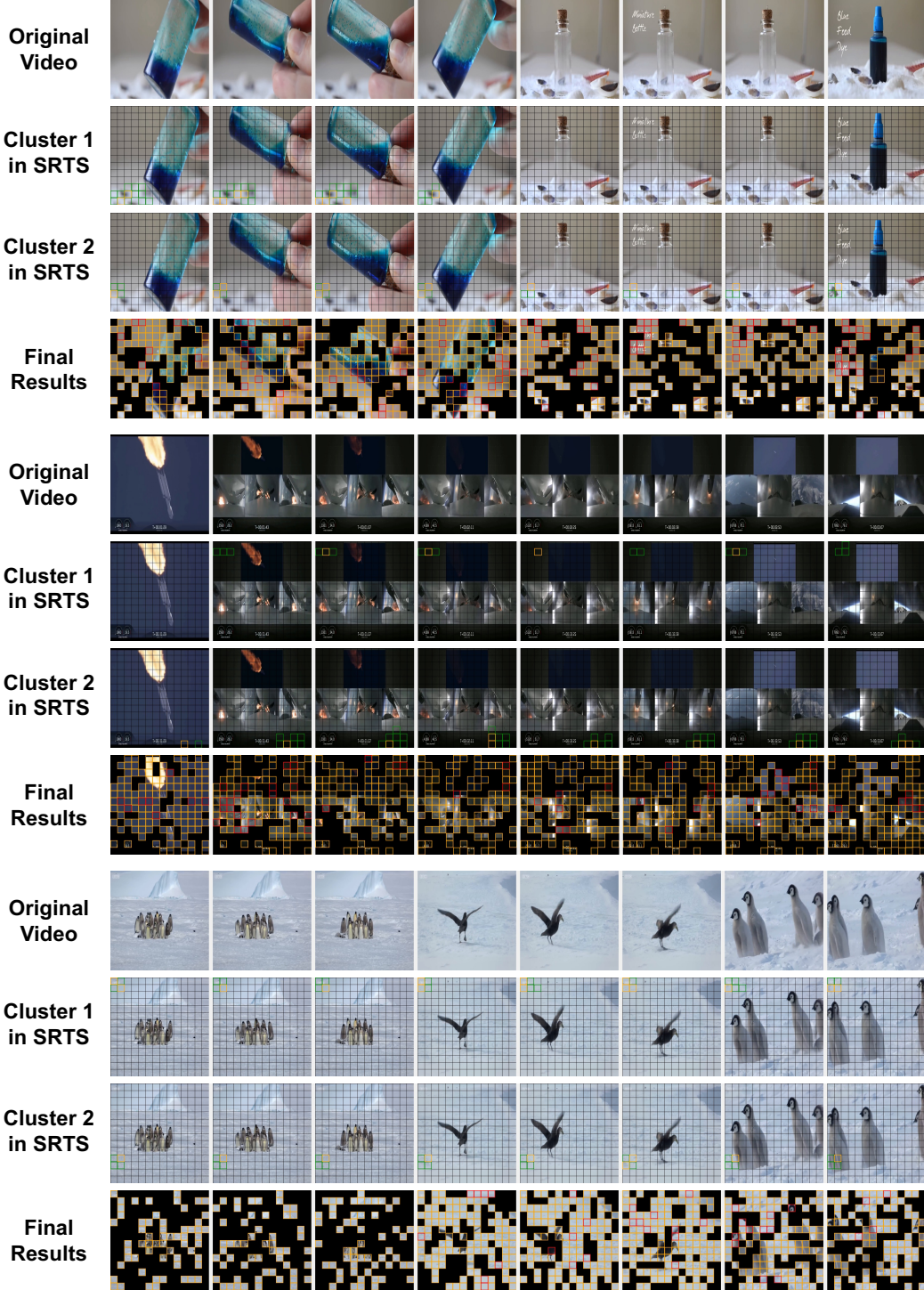


Figure 4: Visualization of the ST-SimDiff contribution in action. SRTS identifies redundant communities (Cluster 1 and Cluster 2) and retains sparse representative tokens for stable content, while DETS marks dense event tokens around dynamic actions.

Figure 4 visualizes the selection process to make the method’s two contributions easier to inspect. The sample sequence contains a dynamic object manipulation task, so it includes both stable visual

regions and event-driving motion. For Similarity-based Representative Token Selection (SRTS), the rows labeled “Cluster 1” and “Cluster 2” show how graph community detection groups spatially and temporally redundant regions, including static background and stationary objects. The sparse yellow boxes in the final results are the retained central representatives, illustrating how SRTS compresses stable content without requiring many background tokens.

The same visualization also makes the DETS contribution visible. In the “Final Results” row, dense red boxes mark tokens selected by high temporal difference. These boxes align with the moving hand and shifting blue liquid, showing that the method locates “turning points” where temporal difference exceeds the threshold (τ_{diff}). DETS therefore preserves fine-grained action details and rapid visual changes that could be underrepresented by representative-token selection alone.

Together, the yellow and red selections summarize the paper’s contribution structure. Sparse yellow tokens represent stable background content, while dense red tokens preserve rapid motion. This overlay provides qualitative evidence for the similarity–difference balance: ST-SimDiff prunes spatiotemporal redundancy while avoiding the blind spot of similarity-only selection during abrupt changes. The visualization therefore supports the main claim that efficiency and event preservation should be optimized together.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Supplementary generalization results on Qwen2.5-VL-7B for 64 and 128 input frames (%). The table tests whether ST-SimDiff retains its efficiency-performance behavior on an additional base model.

To further test the generality of the contribution, we evaluate ST-SimDiff on Qwen2.5-VL in addition to the base models reported in the main experiments. These supplementary experiments use VideoMME and LongVideoBench and examine whether the similarity–difference selection principle remains effective under a different model architecture and with both 64-frame and 128-frame inputs.

Table 4 shows that ST-SimDiff maintains strong performance on Qwen2.5-VL. With a 30% retention ratio, the 64-frame setting reaches 62.4 on VideoMME and 56.8 on LongVideoBench. Increasing the retain ratio to 50% raises performance to 62.9 on VideoMME and 58.4 on LongVideoBench. At this 50% budget, VideoMME matches the corresponding full-token upper bound, and LongVideoBench remains close to full performance. These results reinforce the contribution claim that the framework is not tied to a single LVLM backbone.

C ABLATION STUDY RESULTS ON NVILA

To test whether the component-level contribution transfers across backbones, we repeat the ablation study on NVILA-Video-8B (Liu et al., 2024c). The setup follows the main-paper ablation: a Baseline with fundamental importance-based pruning is augmented first with the similarity selection module (+ Sim) and then with the difference selection module (++ Diff).

Method	VideoMME	LongVideoBench	EgoSchema	
Token Retain Ratio $r = 30\%$				
Baseline	59.3	53.8	50.2	
+ Sim	Spatial	59.5	54.5	50.9
	Temporal	59.6	54.3	51.2
	Spa. + Tem.	59.9	54.6	51.3
++ Diff (Proposed)	60.2	55.2	51.7	
Token Retain Ratio $r = 50\%$				
Baseline	60.9	55.3	51.5	
+ Sim	Spatial	61.1	55.7	51.9
	Temporal	61.3	55.6	51.8
	Spa. + Tem.	61.4	56.0	52.3
++ Diff (Proposed)	61.7	56.5	52.5	

Table 5: Supplementary component ablation on NVILA-Video-8B (%). The table separates the contributions of + Sim and ++ Diff under 30% and 50% retain ratios.

As shown in Table 5, the NVILA trends are consistent with the LLaVA-Video ablation. Adding + Sim improves performance at both 30% and 50% token retain ratios, and the spatio-temporal joint variant (Spa. + Tem.) is stronger than the spatial-only or temporal-only alternatives. For example, at $r = 50\%$, the joint strategy reaches 61.4% on VideoMME, outperforming the other similarity variants. This supports the contribution of joint spatio-temporal redundancy modeling. After the best similarity module is included, adding ++ Diff yields the complete ST-SimDiff framework and provides the strongest results in the table. This supplementary ablation again separates the two roles of the method: + Sim compresses redundant content through representative selection, while ++ Diff preserves changes that define events.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

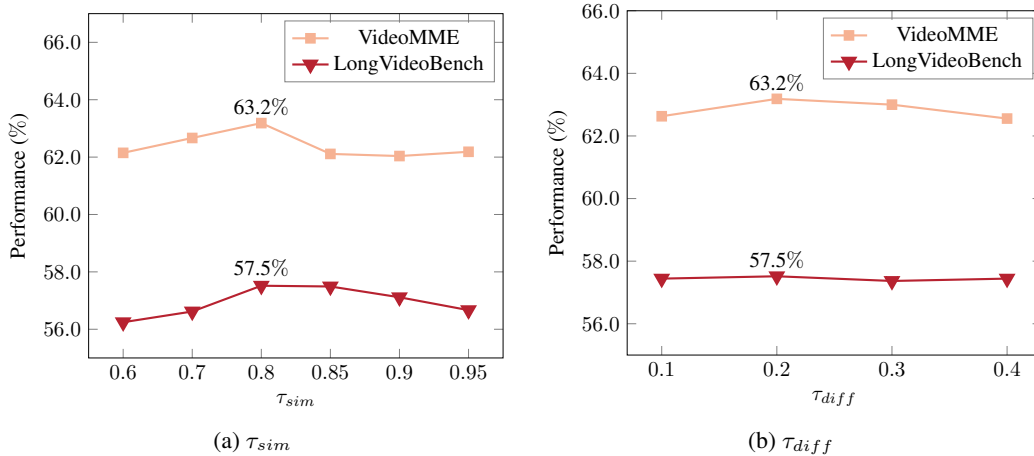


Figure 5: Hyperparameter robustness for the similarity and difference contributions on VideoMME and LongVideoBench. The curves vary τ_{sim} and τ_{diff} while preserving the same evaluation setting.

The similarity threshold τ_{sim} and difference threshold τ_{diff} control the two main contribution paths: representative token selection and transitional event selection. Figure 6 shows that performance first rises and then falls as each threshold changes, while remaining robust within a reasonable range. A very low τ_{sim} makes community detection less precise, whereas a very high value can fragment

semantic clusters. A very low τ_{diff} makes event selection overly sensitive to noise, while a very high value can miss key events. The best observed configuration is $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$, which we use in the other experiments.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

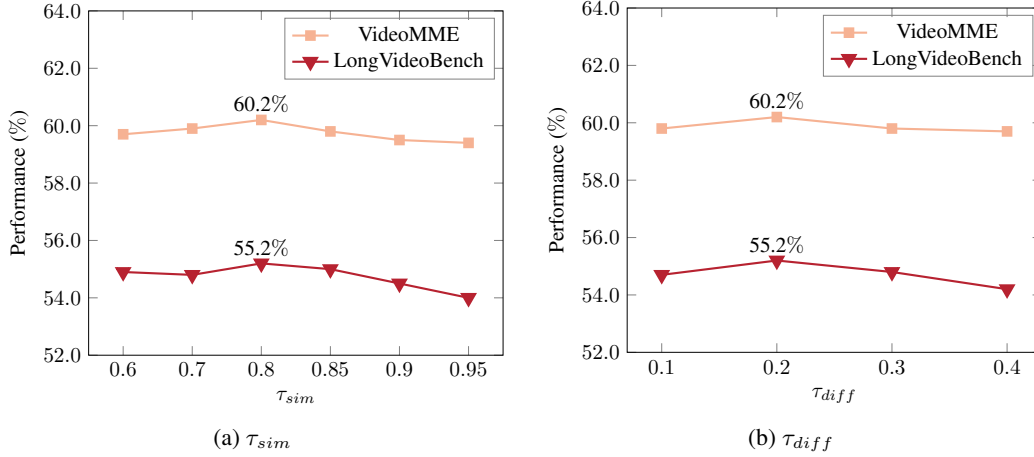


Figure 6: NVILA-Video-8B hyperparameter robustness for τ_{diff} and τ_{sim} . The results test whether the similarity and difference thresholds remain stable on another backbone.

We also evaluate threshold robustness on NVILA-Video-8B to show that the contribution does not depend on a fragile hyperparameter choice. As shown in Fig. 6, we vary the similarity threshold, τ_{sim} , and the difference threshold, τ_{diff} , and measure performance on VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . We test τ_{sim} in the range of $[0.6, 0.95]$ to measure the stability of representative-token selection. Performance is relatively insensitive to this threshold: VideoMME peaks at 60.2% when $\tau_{sim} = 0.8$, and LongVideoBench also peaks at 55.2% at $\tau_{sim} = 0.8$. Although performance decreases slightly when the threshold is too low or too high, the small fluctuation indicates that the similarity-selection contribution is stable across a broad range.

Impact of Difference Threshold τ_{diff} . We test τ_{diff} in the range of $[0.1, 0.4]$ to measure the stability of event-token selection. The method is again robust, with the best performance around $\tau_{diff} = 0.2$: 60.2% on VideoMME and 55.2% on LongVideoBench. This supports the use of a moderate difference threshold for capturing key events while maintaining compression.

In summary, the hyperparameter ablations support the robustness of the two contribution paths. The selected values, $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$, consistently provide the best or strongest observed performance across benchmarks and base models, justifying their use in the main experiments.